

7.6 – Digital Twins in AIoT



What is a Digital Twin?

- Virtual replica of a physical asset or system
- Continuously updated with real-time IoT data

AI-Enhanced Intelligence

- Predict behaviour & detect anomalies
- Simulate scenarios & future conditions

Where Digital Twins Are Used

- Manufacturing & industrial operations
- Energy & utilities management
- Smart buildings & infrastructure

Business Value

- Improve planning & optimization
- Enable predictive decision-making
- Reduce risk & operational costs

Key Aspects of Digital Twin Technology

Real-Time Synchronization

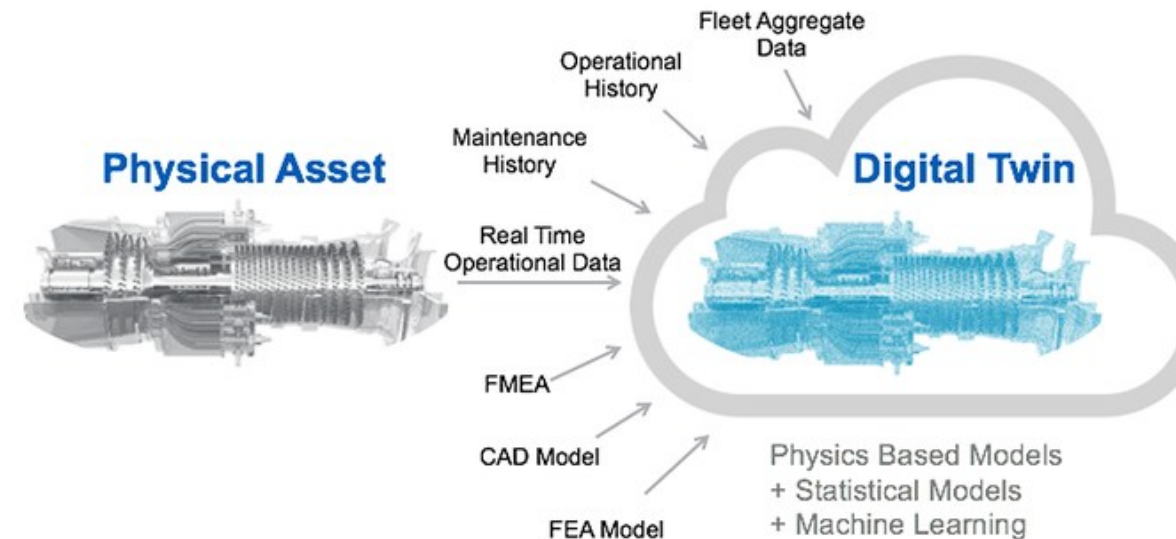
-  Continuously updates using live IoT data
✓ Reflects current conditions, not static models

Core Technology Elements

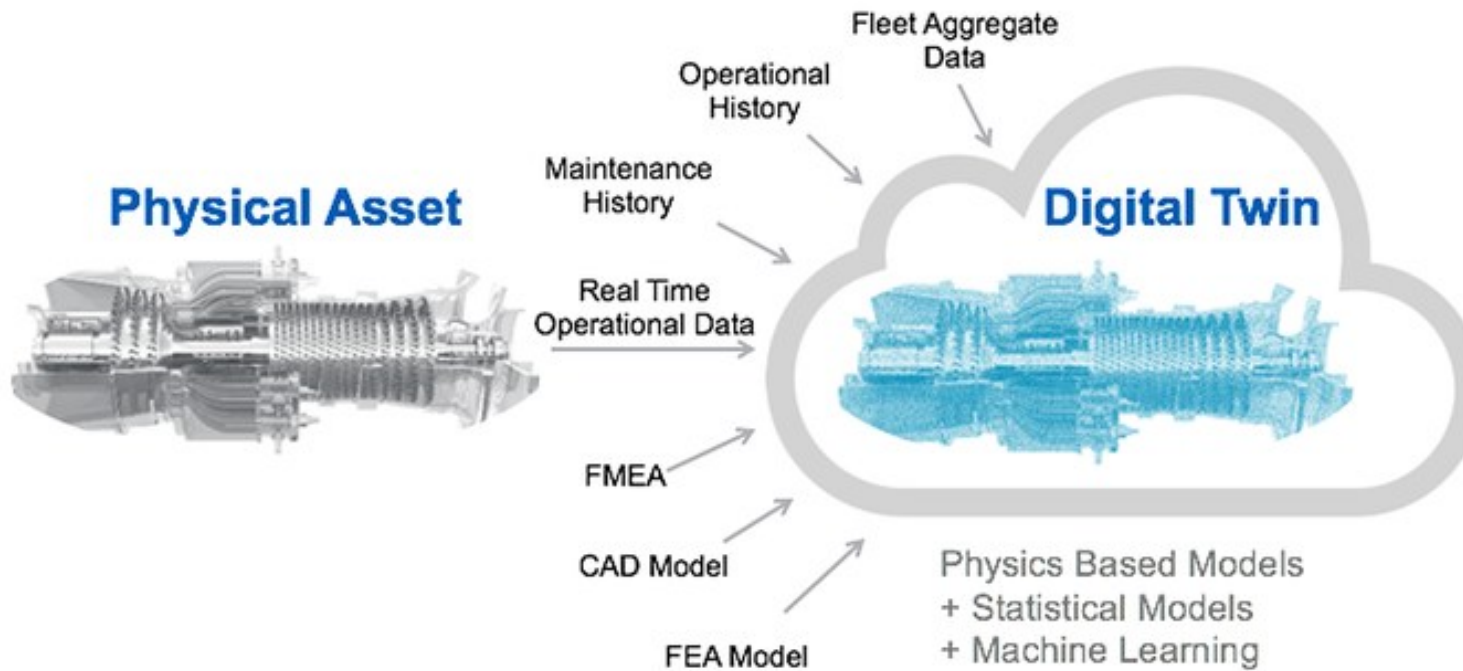
-  3D modeling & visualization
-  IoT-driven real-time data
-  Simulation & predictive analytics

Four Primary Types of Digital Twins

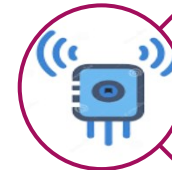
-  **Component Twins** – monitor critical parts
-  **Asset Twins** – represent entire machines
-  **System Twins** – show component interactions
-  **Process Twins** – model complete workflows



Key Components of a Digital Twin System



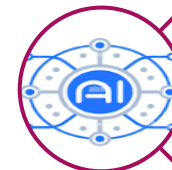
Physical Entity: The actual machine, product, or system.



IoT Sensors: Devices that capture real-time data on performance and conditions.



Virtual Model: A digital representation (3D, simulation) that mirrors the physical counterpart.



AI/Analytics Engine: Algorithms that process data to predict behavior and generate insights.



Data Platform: Cloud or edge infrastructure storing and managing the data.

Case Study: Digital Twin in Wind Turbine Operations



The Problem

Wind farm operators faced multiple operational challenges:

- ▶ Unexpected turbine failures leading to costly downtime
- ▶ Difficulty inspecting turbines in remote or offshore locations
- ▶ Inefficient maintenance schedules (reactive vs predictive)
- ▶ Inability to optimize performance based on changing wind conditions

Unplanned downtime and maintenance costs significantly impacted energy production and profitability.

Case Study: Digital Twin in Wind Turbine Operations

1 Physical System Monitoring

- Sensors installed on turbines to collect:
 - vibration data
 - temperature readings
 - blade stress & rotation speed
 - wind speed & direction

2 Real-Time Data Integration

- Sensor data streamed to cloud platform
- Live operational parameters continuously updated

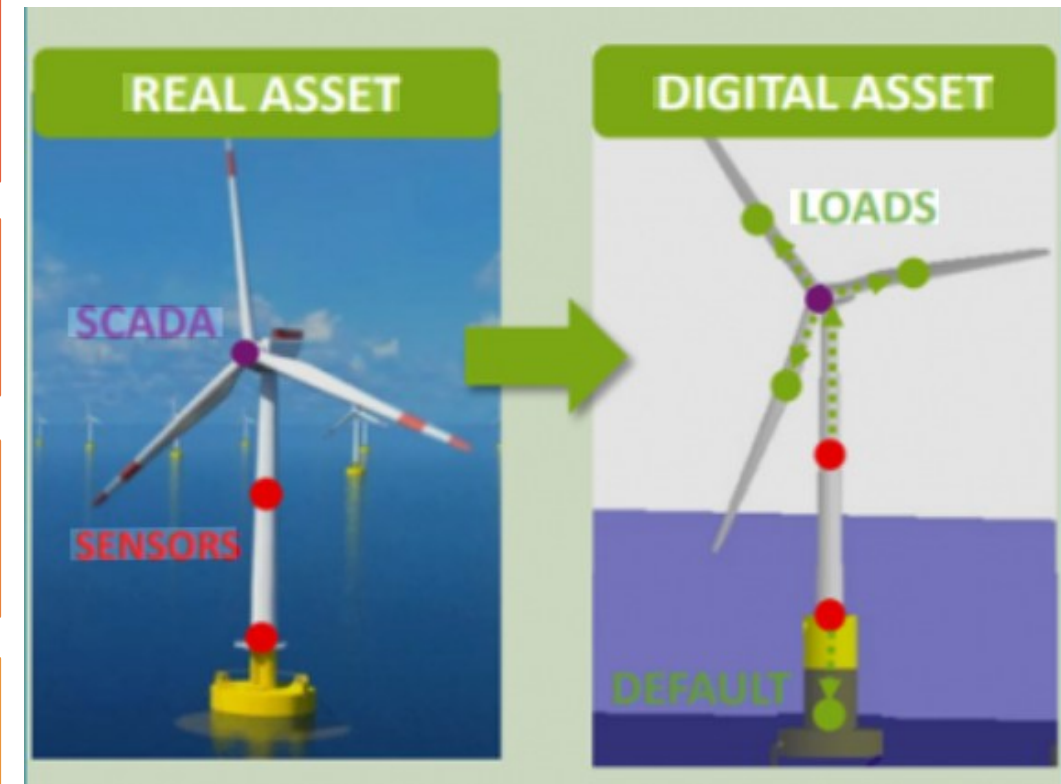
3 Virtual Twin Creation

- 3D digital models created for each turbine
- Real-time performance mirrored in the digital twin

4 AI & Predictive Analytics

- AI models analysed patterns & anomalies
- Simulation tested stress scenarios & weather impacts
- Failure prediction models forecast component wear

Digital Twin Implementation



Case Study: Digital Twin in Wind Turbine Operations

Predictive Maintenance

- ✓ Identified gearbox & blade failures weeks in advance
- ✓ Reduced emergency repairs & downtime

Performance Optimization

- ✓ Adjusted blade pitch & orientation for optimal energy output
- ✓ Improved efficiency under varying wind conditions

Remote Monitoring & Diagnostics

- ✓ Engineers monitored turbines without site visits
- ✓ Reduced maintenance travel & inspection costs

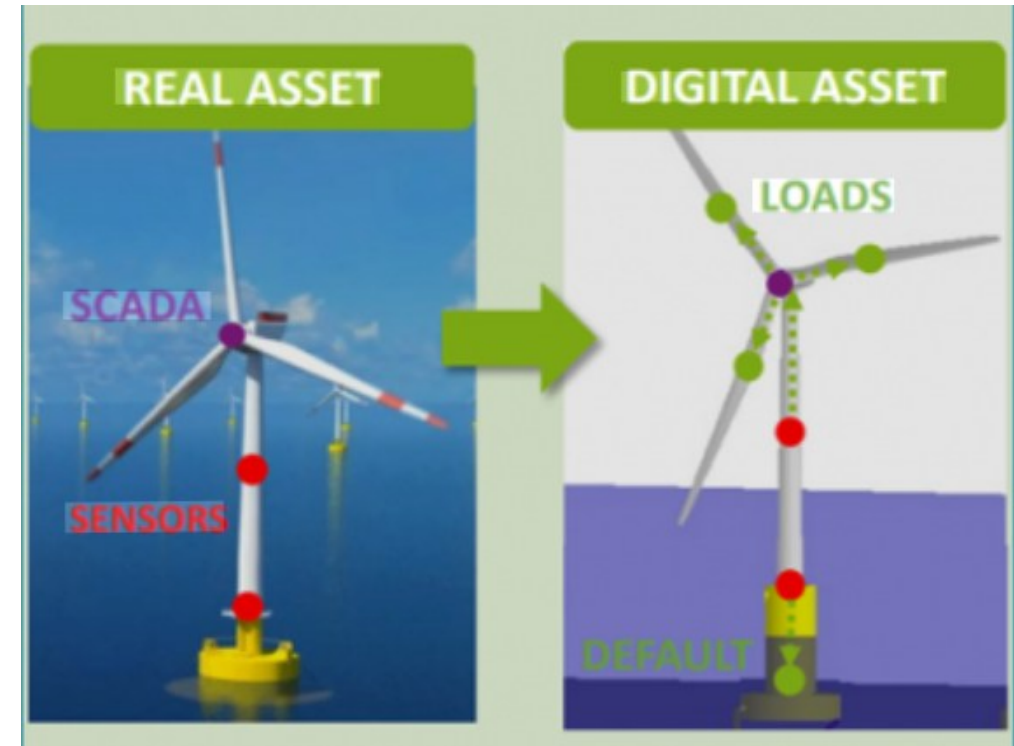
Operational Insights

- ✓ Improved lifecycle planning & asset utilization
- ✓ Increased overall energy generation reliability

Measurable Impact

-  Maintenance costs reduced by up to **25–30%**
-  Energy output improved by **5–10%**
-  Downtime reduced significantly
-  Improved reliability of renewable energy supply

Digital Twin Implementation



Case Study: Digital Twin in Wind Turbine Operations

